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Unlocking Business Value by Introducing Process Cameras

Sobhakar Chanda, Kintali Gupta, Vijay Kumar, Malay Kumar Mandal, Deema Mohammed Alluwaimi, and Shripad Vasant Pandei, ADNOC Offshore, Abu Dhabi, UAE; Harry Thorpe and Paul Stockwell, Process Vision Ltd, Basingstoke, UK

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Abstract

Objective/Scope: This paper describes the initiative to unlock business value through technology deployment i.e. by introducing process cameras and AI models for real-time image processing. Gas compressors require dry gas to operate safely. If liquids or solids enter the compressor, they can cause severe damage, resulting in $ millions in repair costs, production loss, or even catastrophic safety incidents. Traditional sensors often declare gas as ‘dry’ based on calculated hydrocarbon dewpoint, but field experience with process cameras reveals this doesn't always reflect reality.

Process Cameras ensure feed gas quality by detecting liquid contamination early, allowing proactive measures to prevent compressor damage and improve Reliability, Availability and Maintainability (RAM) in gas processing facilities. Using data generated by the process camera, correlating it with process parameters in a Machine Learning (ML) model, bad actors, bottlenecks and opportunities for improvement can be highlighted to improve compliance with an integrity management program in alignment with ASME B31.8S to assist operators in identifying and mitigating fluid-phase threats. Maintenance action, changes in procedure or implementation of a process improvement plan also benefit from evidence-based data from process cameras. It also enables comparison of feed quality between two processing trains and detects upstream performance deterioration by integrating camera technology with conventional instrumentation.

The focus was given to a gas processing platform responsible for 60% of offshore production. The facility has two turbine-driven compressors with two stages each, scrubbers at multiple points, first and second stage coolers, separate gas dehydration systems with coalescer filters and absorbers, and a shared TEG unit. Historically, one compressor bundle fails frequently, leading to million-dollar losses. By installing process cameras, crews can act before contamination occurs.

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Methods, Procedures, Process

High-resolution process cameras capable of operation at high pressure in hazardous area locations were installed to continuously capture video footage of internal gas pipeline conditions at the entry to compressors on a two-train system. Double block and bleed valves allowed the camaras to be isolated for installation and removal while providing a vertical view of pipeline activity from above (Figure 2.). These images were synchronized with process data to create a comprehensive dataset for analysis. The performance of both trains was observed under various flow conditions, including routine operation, periods of low flow, compressor trips, and recirculation events. Particular attention was given to identifying flow patterns such as mist or stratified liquids, as well as how each train responded to identical process inputs. The observations were ultimately correlated with upstream conditions, including pressure, temperature, and flow.

Figure 1—Process Camera installed on the suction side of the stage 1 compressor

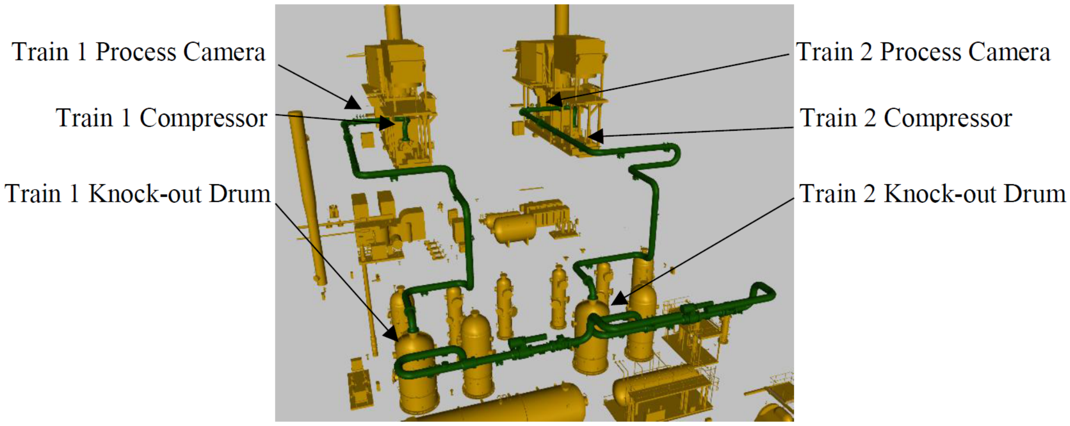


Figure 2—Position of Process Cameras

Results and Observations

When cameras on Train 1 and Train 2 were commissioned, it was immediately clear that there were two very different results. Figure 3 shows the different images. After checking the proper operation of both systems, it was clear that one of the project aims was to better understand these differences. Train 1 shows a significantly darker pipeline than Train 2.

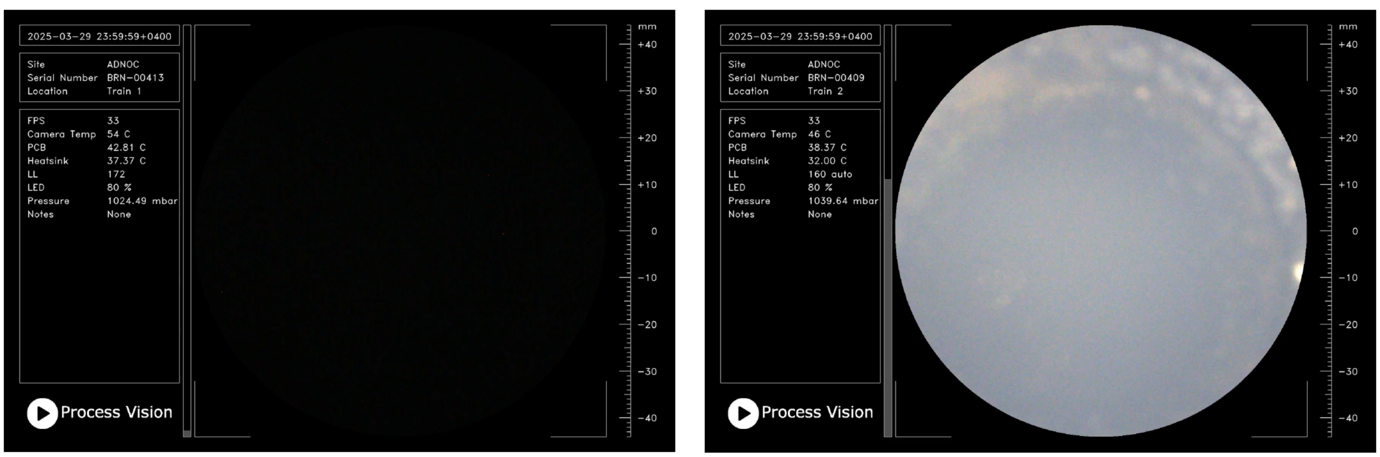


Figure 3—User interface of Train 1 (left) and Train 2 (right)

Data is reviewed by both a human operator and the "meta data" of the images. Meta data consists of three main components for each frame being analyzed:

- Brightness – the sum of the amount of all light received by every pixel within the field of view
- Brightness Variance – how much brightness varies across pixels in the field of view.
- Laplacian Variance – a focus measure used in image processing to quantify how sharp or blurry an image is.

Reviewing meta data reveals that Train 1 showed differences in brightness, in accordance with Train 2 but at a lower level. When compared with three process parameters of Gas Flow Rate, Pressure and Temperature (Figure 4), it can be seen that brightness increases in both trains when the flow rate drops below 1,000 MMSCF/D. It can also be seen that there is a corresponding temperature increase when flow is below 1,000 MMSCF/D. This may be due to recycling gas from the compressor outlet being mixed with the inlet gas stream during low flow conditions to prevent surge and maintain stable operations. This hotter gas has been compressed and would therefore passed its dewpoint and show as an increase in mist level.

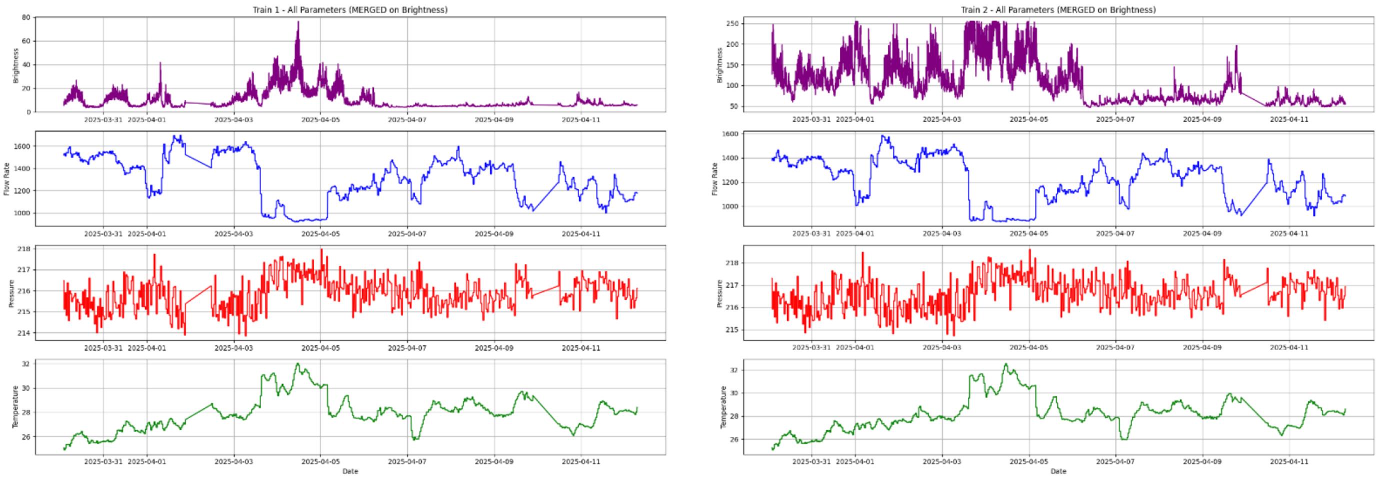


Figure 4—Comparison of brightness under changing process conditions

Machine Learning to Improve the Impact of New Data

A Random Forest Prediction approach was established in a supervised learning model to ultimately provide a digital twin of the processes involved. The model has two aims:

1. Provide better understanding of the process parameters driving the mist flow to provide a diagnostic tool
2. Using flow pressure and temperature, develop a prediction model of mist level as a digital twin to provide "smart alarms". If the live meta data falls outside of the predicted values, an alarm can be activated.

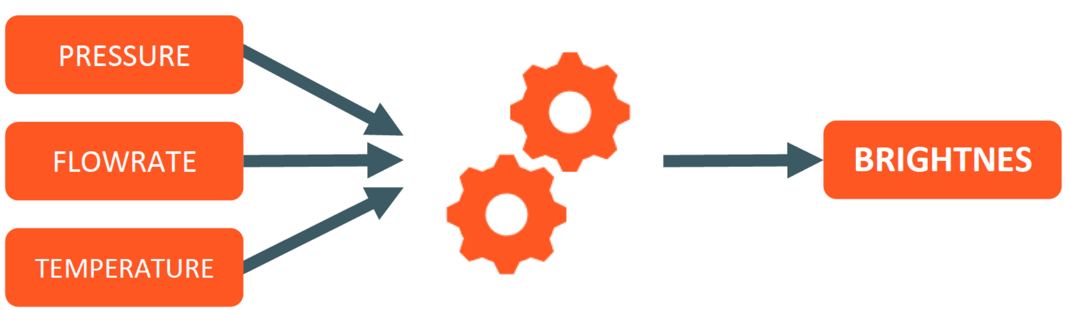


Figure 5—Simplified model of data inputs and outputs to create a digital twin for brightness

Using process parameters as inputs and the three parameters of image meta data as outputs, a machine learning model was established. 15 process parameters including gas flow rate, pressure, temperature, liquid level in knock-out drum and differential pressures at various points in the process were included as inputs. The ML model is completely unbiased; it does not care what the parameters represent only how they relate to the output parameters. In this way some surprising dependencies can be revealed that are not obvious from manual review of the different parameters.

Figure 6 shows the actual brightness versus the predicted brightness for the ML model. Achieving a coefficient of determination (R^2) of 0.92 for both Train 1 and Train 2 (where a perfect match is R^2=1).

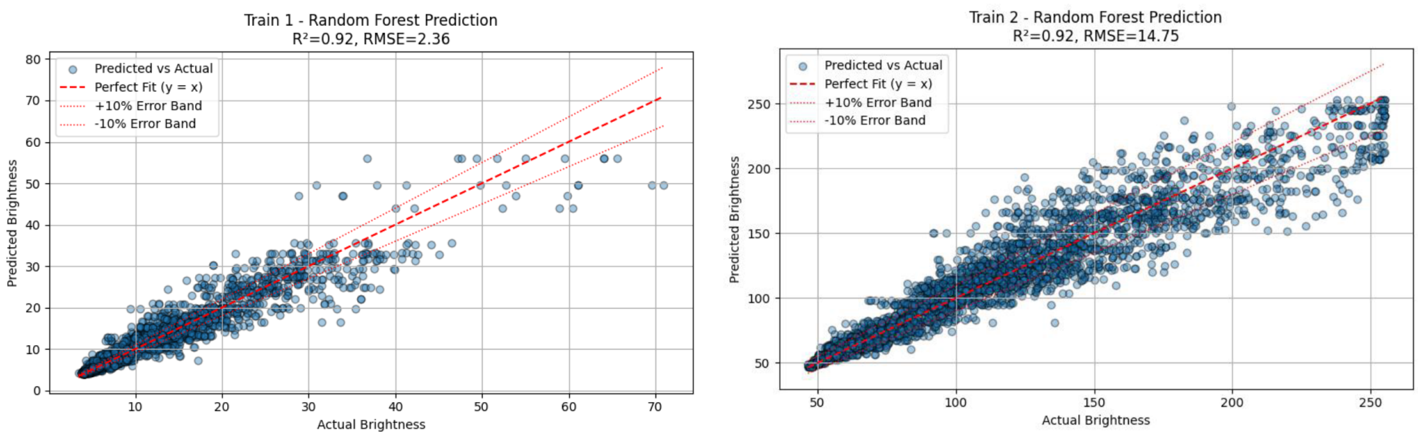


Figure 6—Prediction models for Train 1 and Train 2

Good prediction of mist levels under "normal" conditions of changing flow, temperature and pressure allows this data to be used to provide a "smart alarm". When the meta data falls outside of pre-determined limits, it indicates that something has changed within the process, perhaps a filter needs changing out, a de-mister pad cleaned or a more serious process failure, for example a failure within liquid/gas heat-exchanger allowing liquids into the gas stream. Further work and more data can refine the model.

A Pearson correlation matrix (Figure 7) was generated to provide an initial assessment of linear relationships between individual process parameters and the image meta data from the process cameras. The meta data comprised brightness, brightness variance, and Laplacian variance, computed per frame and aggregated into one-minute statistics, including average, minimum, and maximum values. Process data for the study included flow rate, pressure, temperature, differential pressures, and liquid levels in relevant vessels. In the initial dataset, brightness values were recorded at approximately 54-second intervals and process tags at 36-minute intervals, while later data collection used one-minute brightness and 48-minute process intervals. Alignment onto a common time base was carried out prior to correlation analysis to minimise timing bias.

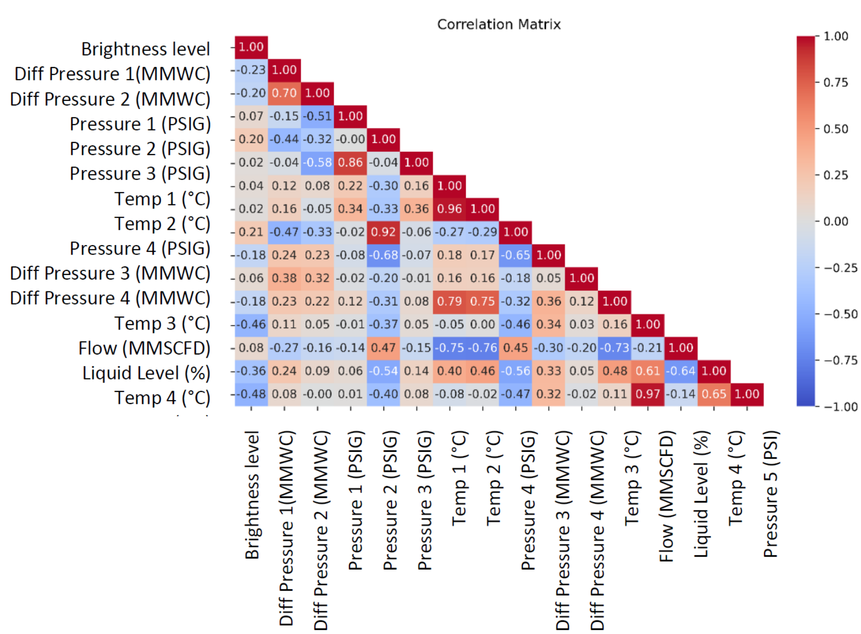


Figure 7—Correlation Matrix for Brightness Vs Process Parameters

The Pearson correlation coefficient quantifies the strength and direction of a linear relationship between two variables, with values ranging from −1 (perfect negative relationship) to +1 (perfect positive relationship). Values close to zero indicate no linear relationship. In this study, the correlation matrix revealed that temperature exhibited the strongest linear relationship to brightness, followed by specific differential pressures and separator liquid level. These findings aligned with process understanding, as all three factors influence the proximity of the gas stream to its dewpoint and therefore the likelihood of mist formation.

The correlation matrix is a useful exploratory tool for several reasons. It provides a rapid means of identifying parameters with potential influence on mist level, highlights variables that may merit closer operational monitoring, and offers an accessible way to communicate initial insights. However, it is subject to several limitations in the context of complex gas processing systems:

1. **Pairwise limitation** – relationships are evaluated one input against one output at a time, without considering interactions between variables.
2. **Linear assumption** – non-linear relationships, common in compression systems, may be underestimated or overlooked.
3. **Multicollinearity sensitivity** – strong correlations between input variables can distort the apparent influence of shared drivers.
4. **Timing sensitivity** – differences in sampling frequency or misalignment between datasets can reduce correlation accuracy.

For these reasons, the Pearson correlation matrix is best regarded as a preliminary screening tool. Subsequent model-based approaches, such as Random Forest feature importance, are more suitable for capturing non-linear effects and variable interactions, providing a deeper and more reliable understanding of parameter influence.

Feature Importance

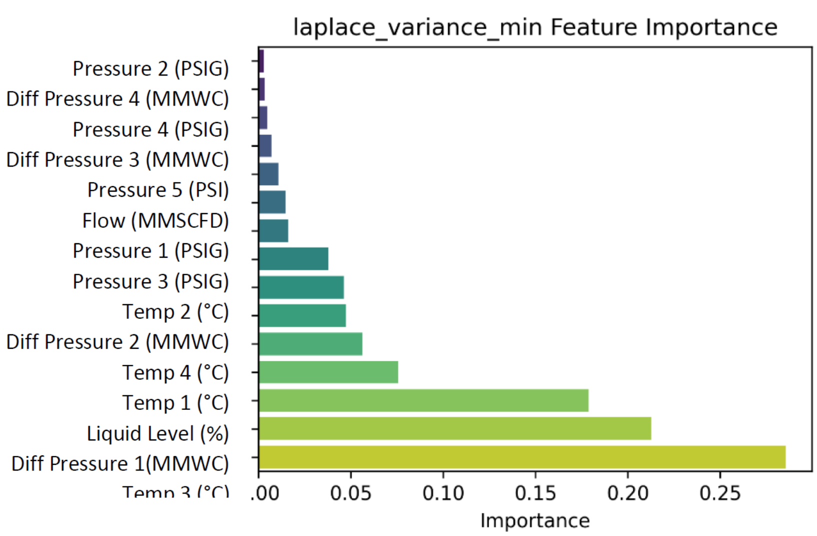


Figure 8—Feature Importance for Laplacian Variance Vs Process Parameters

A model-based feature importance analysis was used to quantify the relative influence of each process parameter on image outputs. The supervised model was a Random Forest regressor trained model to predict camera meta data such as brightness from process inputs that included flow rate, pressure, temperature, differential pressures, and separator liquid level among other tags. The model achieved a coefficient of determination of about 0.92 for both trains, indicating a strong fit to observed brightness behaviour.

Method

Random Forests construct an ensemble of decision trees. At each split a feature is chosen that yields the greatest reduction in prediction error. For regression this is typically the reduction in variance. The cumulative error reduction attributable to a feature across all trees defines its importance score. Normalising these scores yields a ranking that indicates how much each input contributed to accurate predictions in the presence of the other inputs.

Why model-based importance adds value

1. Captures nonlinear relationships between process parameters and mist indicators that are not visible to linear correlation.
2. Accounts for interactions between inputs. If temperature and a specific differential pressure act together, their joint effect is reflected in split choices across trees.
3. Mitigates the impact of multicollinearity by distributing importance across correlated tags according to their marginal contribution to error reduction.
4. Aligns directly to the prediction objective. Importance is measured with respect to reducing brightness prediction error rather than pairwise association.

Interpreting Results for Operations

In this study, feature importance confirmed temperature as a dominant driver of brightness and Laplacian variance and highlighted additional contributors that were less prominent in simple correlation views, such as selected differential pressures and vessel liquid levels. These rankings guided three practical actions. First, focus routine checks on the highest impact tags and their transmitters. Second, tune alarm logic and thresholds on variables that materially shift predicted brightness. Third, prioritise improvement projects that influence the most important drivers, for example separator control and recycle strategies.

Position in the Workflow

Feature importance complements the Pearson correlation matrix. The correlation matrix provides an accessible first look. Model based importance then provides a more reliable basis for decision making in a complex system where effects are nonlinear and interactive. The combination supports both communications to plant teams and the development of robust smart alarms that compare live meta data against model predictions.

Observations & Data

Debris

Close examination of the recorded footage revealed two brief but notable instances where coloured debris passed through the camera's field of view.

As illustrated in Figure 9, these fragments exhibited a distinct orange or rust-like hue, clearly different from the surrounding greyscale tones of the gas flow. The colouring strongly suggests that the particles could be flakes of protective coating or corrosion products such as iron oxide being shed from upstream equipment, possibly because of erosion, mechanical disturbance, or chemical attack. While these events were momentary, their presence is significant — they point to the possibility of developing integrity issues that may not be visible through conventional instrumentation.

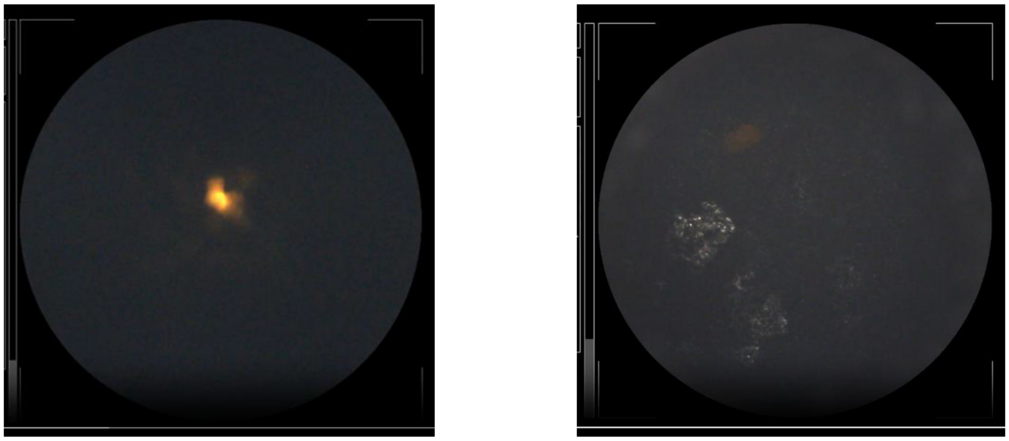


Figure 9—Debris flying past in the gas flow

Because the process camera captures in full colour — even though most flow activity appears within the greyscale range — it is feasible to implement an image-processing routine that automatically scans each frame for specific colour signatures. When such a signature is detected, the system could immediately flag and store the relevant frames, along with a timestamp, for further engineering review. Over time, this data could help operators track the frequency, size, and nature of such debris events, building an early-warning picture of upstream corrosion or coating failure.

This capability transforms the camera from a passive observer into an active diagnostic tool, supporting predictive maintenance strategies and aligning with asset integrity management best practices. Detecting the onset of corrosion at such an early stage can avoid far greater downstream costs, protect sensitive equipment such as compressors, and help maintain gas quality standards at custody transfer points.

Problem – Detection – Action Call-Out Box		
Problem	**Detection**	**Action**
Upstream corrosion or coating degradation releases particulate debris into the gas stream.	Process camera detects color anomalies (e.g., orange/rust tones) using automated image processing.	Flag and log events; investigate upstream assets; schedule maintenance or targeted inspection to prevent escalation.

Plant Trip Events

During the study period, two plant trips occurred and were able to be analyzed from a pipeline activity viewpoint. On both occasions a sudden pressure change saw an immediate high level of mist as the gas hit and surpassed its dewpoint. When the mist cleared significant levels of liquid were observed on the pipe wall (Figure 10).

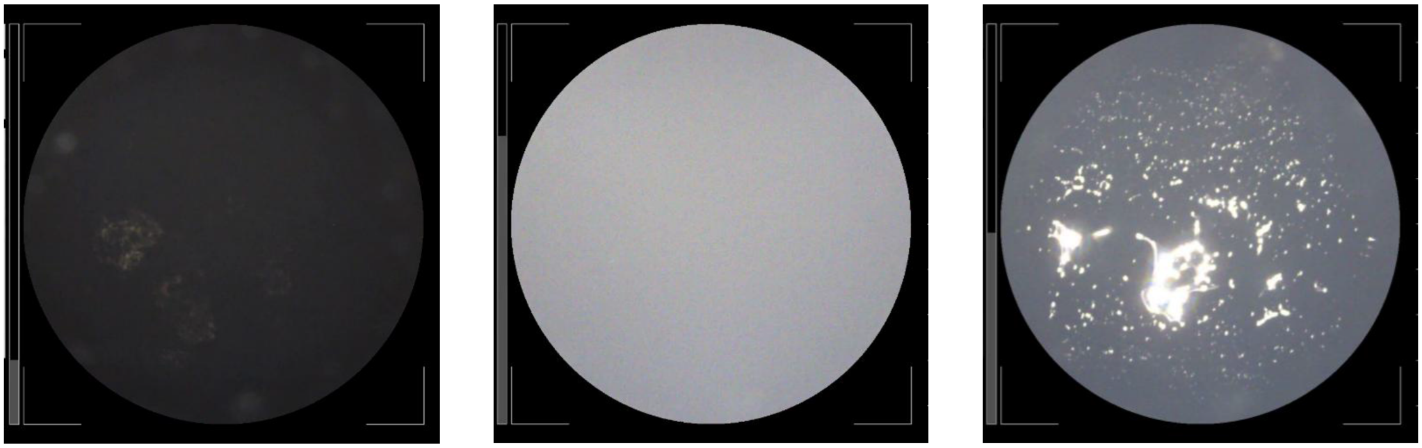


Figure 10—Left image shows pipeline activity before the plant trip. The middle shows sudden mist at the plant trip. The right shows liquids condense on the pipe wall as the mist clears.

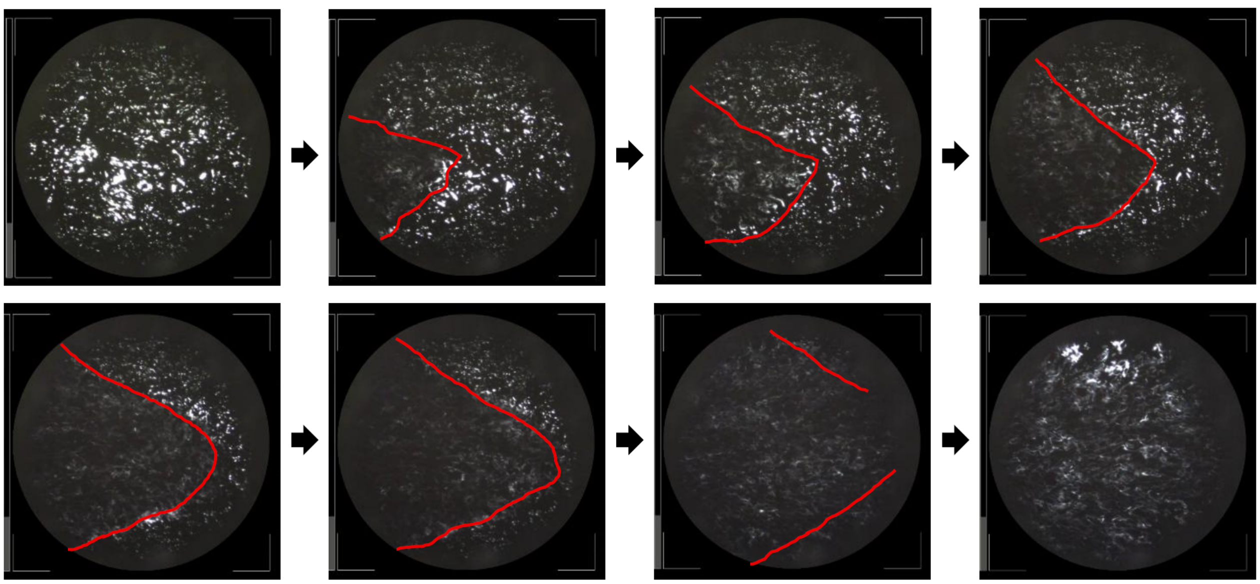


Figure 11—Sequential frames of liquid flowing into the compressor

As the plant came back online, real time footage at the re-start of the compressors was analyzed and sequential frames show the progress of the liquid flow, highlighted in red, directly entering the compressor.

The risk of allowing liquids to enter gas compressors should not be understated. Should sufficient liquids enter the compressor, a compressor trip occurs or a catastrophic event. Relying on the last line of defense (the compressor trip) is not good operational practice and endangers personnel local to the compressor. In addition, early detection and prevention of liquids in either mist flow or stratified flow can extend the average dry-gas-seal life of dry gas seals for roughly one year to more than three years in natural gas compressors, significantly decreasing unplanned outages, servicing costs and seal-gas consumption. With liquids often being present at start-up, operators can use live video to either de-pressurize the section of pipe to vaporize the liquids or hold off flow ramp-up until the liquids have moved through the compressor. Risks can be better evaluated for personal safety and compressor trips or worse can be prevented.

Adding Business Value Across the Supply Chain

There are applications for this technology across the supply chain. For both gas suppliers and gas receivers, this visibility presents a chance to significantly improve commercial outcomes:

- **For Gas Suppliers**, detecting and managing liquid carryovers at the inlet to gas processing can reduce the risk of foaming and fouling within the de-sulfurization and de-humidification processes. At the processing export line, lowering condensate carryover means maximizing revenue by ensuring valuable NGLs are recovered and sold rather than passing through custody meters unnoticed. Even a small liquid volume fraction, just 0.1% in a 100 MMSCFD export line at 1,000 psi can equate to more than 748 gallons/day of saleable NGLs. At a conservative $51.3/bbl, it represents over $5.6 million/year in potential revenue.
- **For Gas Receivers**, early identification of liquids in the gas stream reduces downstream maintenance costs, mitigates corrosion and erosion risk, and optimizes pigging schedules. Avoiding unnecessary pig runs (often ~$35k/mile) and unplanned compressor servicing directly protects the bottom line by $millions/year while improving operational RAM.

The ability to *see* what's flowing offers a win–win: greater profitability, enhanced safety margins, and better-informed operational decisions for all stakeholders.

Gas-Processing Trains

Live video footage allows operators to balance parallel separators and run closer to optimum throughput. With tighter control on liquid carryover at the front end of gas processing, plants previously limited by fear of foaming can boost production by several percentage points of capacity because they can now watch, not guess, for liquid breakthroughs.

When process cameras are installed on the export gas line improved NGL recovery can be implemented and demonstrated increasing the plant's condensate revenue stream.

Custody-Transfer Metering

A pipeline operator used video evidence of stratified NGLs to escalate a gas quality problem with a gas supplier; the dispute was quickly resolved with footage from a process camera on the supplier side showing that the liquid event was transient and that the event was over on the supply side. Thereby preventing valve closure or other action by the pipeline operator.

Power Stations

By the time natural gas reaches a turbine power station, it may carry unwanted contaminants, including liquids from upstream suppliers, compressor oil carryover, and iron sulfide from pipe walls. If not effectively removed, these can cause significant damage. Video footage taken downstream of filter packs has revealed filter breakthrough under varying operating conditions, a problem that differential pressure (DP) readings can miss, since DP shows what the filter has already stopped.

With better visibility of what passes through, operators can fine-tune inlet gas heater control, prevent fuel nozzle blockages and reduce hot-spot formation, ultimately improving turbine performance and reliability.

LNG Feed Gas

One study[1] showed that just 3PPB of heavy hydrocarbons (C12+) is sufficient to block a cold box in just 30 days. With many LNG plants having multiple supply lines, knowing which these suppliers are delivering liquids is vital for continued, uninterrupted LNG production and to ensure sufficient and effective filtration and separation system continue to perform throughout their design life.

Compromised Cathodic Protection Systems

Asset integrity management programs, aligned with ASME B31.8S-2002, can use liquid carryover detection to identify heightened risks for feedback into mitigation actions to compressors and other critical infrastructure. Stratified flows often transport electrically conductive solids that can bridge isolating joints, weakening or bypassing cathodic protection systems. This increases the likelihood of corrosion and creates a slow, often unnoticed, pathway for methane release.

Early detection allows integrity teams to intervene with targeted pigging before the issue escalates into a reportable incident. Helping operators protect infrastructure, maintain compliance, and meet methane-intensity reduction commitments demanded by investors and regulators worldwide.

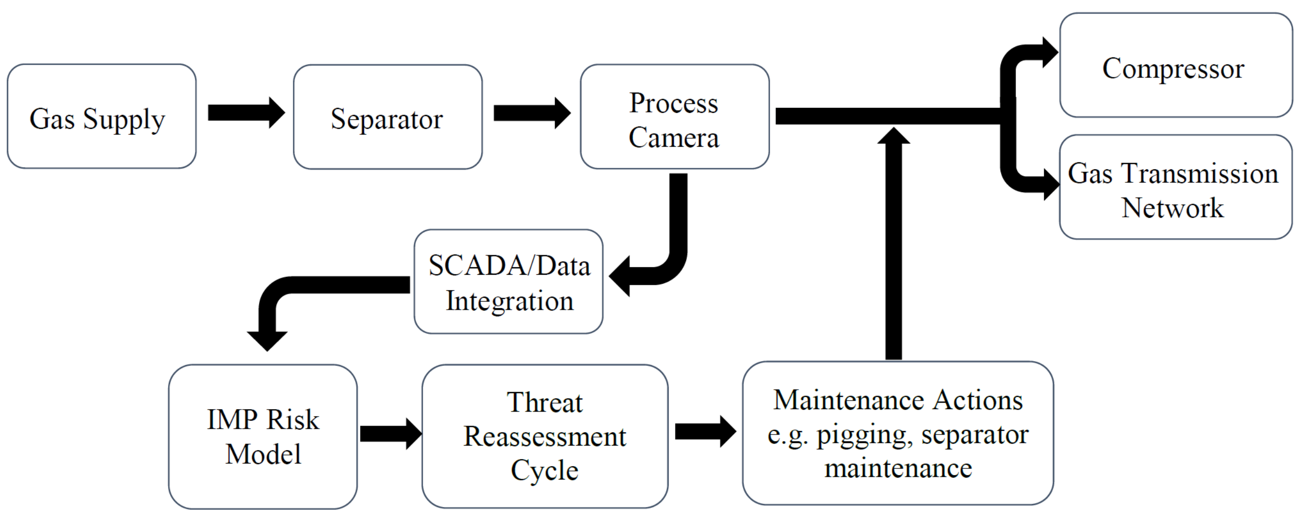
A Broader Perspective on Digital Transformation

Cameras, analytics, and cloud-based reporting may sound cutting-edge, but they follow the same trajectory already well established in other industries. From predictive maintenance on rotating machinery, to satellite monitoring of pipeline corridors, to drone inspections of flare stacks. Process Cameras apply this same digital transformation to process control, much like CCTV transformed the security industry.

What sets process cameras apart is their ability to connect directly to existing tapping points with minimal installation effort, while producing a very small data footprint. Cybersecurity, often a concern with cloud-connected systems, has been built in from the start, with secure global access, remote software updates, and continuous system health monitoring ensuring both operational resilience and data protection.

Integration Into Integrity Management Programs

ASME B31.8S-2022 requires operators to identify and mitigate all threats, including fluid-phase issues. Liquid carryover is a well-recognized integrity threat that can lead to asset damage if unmanaged. Process cameras provide real-time visibility and alarm capabilities. Liquid carryover incident reports can be used to feed data into the Integrity Management Program (IMP) risk management process. Integrating liquid carryover monitoring into the IMP framework strengthens regulatory compliance, reduces unplanned downtime, and mitigates financial risk from equipment failures.



Conclusions

This study illustrates the substantial benefits of visual monitoring in gas processing environments. Providing real-time views of upstream process facility performance, such as Knock-Out Drums and De-liquidizers, enhanced decision-making regarding filter maintenance and system integrity. Machine learning techniques were utilized to analyze video metadata, including brightness levels, to correlate visual flow regimes with process variables. This enabled predictive alarms and offline modelling. Sensitivity analysis identified key drivers of mist flow appearance, thereby creating new opportunities for process optimization. This integrated approach marks a pivotal step towards developing intelligent systems.

The commercial benefits are significant. With compressor damage and trips costing up to $10 million in repairs and downtime. Even a single avoided trip valued at approximately $600,000. The improvement of safety and cost recovery can easily justify an investment in multiple camera systems. These camera systems transform invisible risks into visible opportunities for safer, more efficient operations. In addition, the gains from improved management of liquid carryover in terms of process efficiency gains provides justification for rapid ROI.

Benefits

- Real-time visibility of actual gas conditions—not inferred values
- Enhanced safety and informed decisions at start-up and ramp-up
- Reduced unplanned downtime and maintenance costs
- Better understanding of system behavior and improved asset management
- Demonstrates improved compliance with ASME B31.8S-2022 and CFR Part 192
- Predictive insights using AI and visual data correlation

Way forward

- This trial mainly focused on comparing the first stage inlet conditions with observations of both trains. However, there are plans to install additional process cameras within compressor trains, such as at the second stage suction inlet and the gas inlet towards the Glycol absorber, to monitor and help prevent liquid migration to other parts of the compressor trains.
- There are several gas turbines for power generation and gas compression in offshore facility with an history of frequent outages and these failures are primarily attributed to liquid migration toward the burners. To address this issue, there are plans to install process cameras for early detection and timely mitigation, ensuring that the turbines are not operated under adverse conditions for extended periods.

References

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